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Problem statement
We are exploring interventions on how to speed up learning of long-context capabilities in LLMs. Mainly we will focus on training large models for very few steps and distilling into a smaller model. Given limited compute we will focus on training smaller-scale models on synthetic tasks

Motivation
Long context task performance is one of the prominent challenges in training a capable LLM. There are both efficiency concerns (training on long-context data requires much more compute) and performance concerns (model quality is known to degrade at longer context lengths).

Guiding question or hypothesis
Hypothesis: larger models with more parameters learn long context tasks more efficiently
Question: does model scale consistently correlate with how efficient a model is at learning long-context tasks? and is there a point where distilling from a much larger model is more efficient than only training a small model?

Initial related work
Lost in the Middle: How Language Models Use Long Contexts
> establishes the motivation
https://arxiv.org/abs/2307.03172

Emergent Capabilities Arise Randomly from Learning Sparse Attention Patterns
> measures long-context inefficiency on synthetic tasks; proposes interventions like distillation
https://arxiv.org/abs/2606.25010

Possible datasets and/or resources
https://database.lichess.org/ - we could explore training small models on chess, and observe how performance degrades with longer chess games